

Overhead-Line-Mechanical-Clearance-Tool

Technical Documentation

Inputs · Equations · Calculation Flow · Outputs

Standards IEC 60826:2017 IEEE 738-2023 EN 50341 ICAO Std Atmosphere	Scope Mechanical wind & sag Thermal ampacity Cable state- change EEHC conductor DB	Author / Platform Mohamed Ezaldine Windows EXE + Python 3 Custom Kinter · Matplotlib
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1. Overview

Overhead-Line-Mechanical-Clearance-Tool is a desktop engineering application for bare overhead AL3 conductors. Three coupled calculation modules cover the full mechanical and thermal design workflow:

- **Wind Load** *IEC 60826:2017 3-A* — corrected wind force per meter of conductor.
- **Cable Mechanics** *IEC 60826 / EN 50341* — cubic state-change equation for tension, sag, and swing.
- **Thermal Analysis** *IEEE 738-2023* — steady-state conductor temperature and maximum ampacity.

1.1 Repository Files

File	Description
OverheadLineAnalyzer.exe	Standalone Windows executable — no Python install required
Equations_Reference.pdf	Derivations and explanations for all 17 equations
Technical_Documentation.docx	This file — inputs, equations, and outputs reference
README.md	GitHub repository readme with screenshots

2. Inputs

2.1 Conductor Selection

Input	Unit	Description
Conductor type	—	Drop-down — 13 AL3 conductors 16–400 mm². Selecting auto-fills diameter, weight, RTS, E, DC resistance, and CCC, and suggests a suitable EDS tension and span.

2.2 Mechanical Inputs (IEC 60826)

Input	Unit	Description
Span Length L	m	Horizontal distance between support poles. Valid range 20–500 m. Parabolic sag formula (EQ 6) accurate up to 500 m when sag < 10% of span.
EDS Tension T ₁	kgf	Every-Day Stress: conductor tension at 40 °C under no wind. Baseline for the state-change cubic (EQ 9). Recommended 15–25% of RTS.
Load Case	—	EDS (40 °C, no wind) Max Sag (90 °C, no wind) Max Swing (–10 °C + wind). Each case sets t ₂ and w ₂ for EQ 9.

2.3 Wind Load Inputs (IEC 60826 3-A)

Input	Unit	Description
Wind input mode	—	Toggle: wind speed entry (m/s) OR direct dynamic pressure (kgf/m²). Both converge to q ₀ for EQ 4.
Reference Wind Speed V	m/s	10-minute mean speed at 10 m height (IEC V_RB). Converted to q ₀ via EQ 1. Typical Egypt: 24–36 m/s.
Direct Pressure q ₀	kgf/m²	Supply q ₀ directly from a wind map or national code. Used unchanged in EQ 4.
Site Altitude H	m	Elevation above sea level. ICAO density correction (EQ 2). EEHC: H ≤ 1000 M.
Drag Coefficient C _{xc}	—	Aerodynamic drag of cross-section. IEC 60826 3.3: 1.0 for all round AL3 conductors.
Combined Wind Factor G _c	—	Gust + terrain turbulence factor (IEC Table B.1). Range 1.0–1.7. Urban ≈ 1.3, open terrain ≈ 1.5.
Span Factor G _L	—	Reduces wind load for long spans (EQ 3). Auto calculated or manual. Range 0.85–1.00.
Wind Angle Ω	°	Angle between wind and conductor axis. Ω = 90° (perpendicular) is worst-case, sin² (90°) = 1.

2.4 Thermal Inputs (IEEE 738-2023)

Input	Unit	Description
Operating Current I	A	Actual line current. Bisection solver on EQ 11 finds conductor temperature. Auto-set to conductor CCC.
Wind Speed V_w	m/s	Wind for thermal convection (independent of mechanical wind). IEEE 738 default 0.6 m/s.
Ambient Temperature T_a	°C	Air temperature. EEHC: daily means 35 C, maximum 50 °C.
Emissivity ε	—	Surface radiance efficiency (EQ 15). New Al ε ≈ 0.5; weathered ε ≈ 0.8–0.9.
Solar Absorptivity α_s	—	Fraction of sunlight absorbed (EQ 16). New ≈ 0.5; weathered/dark ≈ 0.9.
Solar Irradiance Q_{se}	W/m ²	Total incident solar radiation. Egypt: 900–1100 W/m ² . Default 1000 W/m ² .

3. Mathematical Equations

3.1 Wind Load Module — IEC 60826:2017

EQ 1 IEC 60826	$q_0 = \frac{1}{2}\rho V^2$	Dynamic wind pressure [N/m ²]
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Air density ρ from EQ 2. Divide result by 9.81 for kgf/m². V = reference wind speed [m/s].

EQ 2 ICAO Std Atm	$\rho = \rho_0 \times (1 - 2.2558 \times 10^{-5} H)^{5.2561}$	Air density vs altitude [kg/m ³]
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$\rho_0 = 1.225 \text{ kg/m}^3$ at sea level, 15 °C. Constants: $2.2558 \times 10^{-5} = L/T_0$; exponent 5.2561 = $g/(L \cdot R)$ from ICAO troposphere model. Valid $H \leq 11\,000 \text{ m}$ (EEHC $\leq 1000 \text{ m}$).

EQ 3 — Span Reduction Factor G_L (IEC 60826 Annex B)

Wind does not blow uniformly along a long span simultaneously. G_L reduces the design wind load for longer spans:

EQ 3 IEC 60826 Annex B	$G_L = 1.00$ when $L \leq 100 \text{ m}$ $G_L = 1.00 - 0.15 \times (L - 100) / 300$ when $100 < L \leq 400 \text{ m}$ $G_L = 0.85$ when $L > 400 \text{ m}$
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EQ 4
IEC 60826

$$w_{\text{wind}} = q_0 \cdot C_{xc} \cdot G_c \cdot G_L \cdot d \cdot \sin^2(\Omega)$$

Wind load per meter [kgf/m]

Symbol	Unit	Meaning
q_0	kgf/m ²	Reference dynamic wind pressure (EQ 1 ÷ 9.81)
C_{xc}	—	Drag coefficient — 1.0 for all round AL3 conductors
G_c	—	Combined wind factor: gust + turbulence (range 1.0–1.7)
G_L	—	Span reduction factor from EQ 3 (range 0.85–1.00)
d	m	Conductor outer diameter from datasheet
$\sin^2(\Omega)$	—	Wind angle factor — 1.0 for perpendicular wind (worst case)

3.2 Cable Mechanics Module — IEC 60826 / EN 50341

EQ 5
IEC 60826

$$w_{\text{result}} = \sqrt{(w_c^2 + w_{\text{wind}}^2)}$$

Resultant load [kgf/m]

Conductor self-weight (vertical) + wind load (horizontal) combined by Pythagoras. Used as w_2 in EQ 9 for the Max Swing case.

EQ 6
Cable Theory

$$\text{sag} = w L^2 / (8 T)$$

Parabolic sag at mid-span [m]

Valid for sag < 10% of span, $L \leq 500$ m. Higher temperature → lower tension T → larger sag. $w = w_c$ (EDS/Max Sag) or w_{result} (Max Swing).

EQ 7
IEC 60826

$$\theta = \arctan(w_{\text{wind}} / w_c)$$

Swing angle from vertical [°]

Angle at which the conductor deviates from vertical under lateral wind. Used to split total sag into ground and structure clearance components.

EQ 8 — Sag Components (IEC 60826 §3)

EQ 8
IEC 60826

$$\text{Vertical sag} = \text{sag} \times \cos(\theta)$$

$$\text{Horizontal swing} = \text{sag} \times \sin(\theta)$$

EQ 9 — State-Change Equation (EN 50341 / Cable Mechanics)

Given EDS tension T_1 at 40 °C, the cubic predicts new tension T_2 after any temperature or load change — accounting for mechanical and thermal elongation simultaneously:

EQ 9 EN 50341	$T_2^3 - K \cdot T_2^2 - M = 0$
	$K = T_1 - (w_1^2 L^2 EA) / (24 T_1^2) - \alpha EA (t_2 - t_1)$
	$M = w_2^2 L^2 EA / 24$
	$f(T_2) = T_2^3 - K \cdot T_2^2 - M \quad f'(T_2) = 3T_2^2 - 2K \cdot T_2$

Newton-Raphson solver: starts at $T_2 = T_1$, iterates $T_{2_new} = T_2 - f(T_2)/f'(T_2)$. Stops when $|\Delta T_2| < 0.001$ kgf or after 200 iterations.

EQ 10 EEHC	$SF = T_2 / RTS \leq 0.95$	Mechanical safety factor
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T_2 must not exceed 95% of the Rated Tensile Strength (RTS). The application raises a **WARNING** and turns the plot title red if this limit is exceeded.

3.3 Thermal Analysis Module — IEEE 738-2023

EQ 11 IEEE 738-2023	$q_c + q_r = q_s + I^2 R(T)$	Steady-state heat balance [W/m]
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Master thermal equation: heat lost (convection + radiation) = heat gained (solar + Joule). The solver finds temperature T where this balance holds exactly.

EQ 12 IEEE 738-2023	$R(T) = R_{20} \times [1 + \alpha_R (T - 20)]$	Resistance at temperature [Ω/m]
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$\alpha_R = 0.00403$ /°C for aluminum. Higher temperature → higher resistance → more Joule heating at same current — a positive feedback loop requiring iterative solution.

EQ 13 — Air Properties at Film Temperature (IEEE 738-2023)

EQ 13 IEEE 738-2023	$T_f = (T_c + T_a) / 2$
	$\rho_f = 1.293 / (1 + 0.00367 \times T_f)$
	$\mu_f = 1.458 \times 10^{-6} \times (T_f + 273)^{1.5} / (T_f + 383.4)$
	$K_f = 0.0242 + 7.2 \times 10^{-5} \times T_f$

Notes: μ_f uses Sutherland's law for air ($383.4 = 273 + 110.4$ K Sutherland constant). K_f is a linear fit valid 0–120 °C. ρ_f uses the ideal gas law ($\rho_0 = 1.293$ kg/m³ at 0 °C).

EQ 14 — Convective Heat Loss (IEEE 738-2023 §4-A)

EQ 14 IEEE 738-2023	$Re = D \times V_w \times \rho_f / \mu_f$
	$q_{cn} = 0.0205 \times \rho_f^{0.5} \times D^{0.75} \times \Delta T^{1.25}$
	$q_{c1} = (1.01 + 1.35 \times Re^{0.52}) \times K_f \times \Delta T$
	$q_{c2} = 0.754 \times Re^{0.6} \times K_f \times \Delta T$
	$q_c = \text{MAX} (q_{cn} , q_{c1} , q_{c2})$

$\Delta T = T_c - T_a$. q_{cn} applies with no wind (natural convection). q_{c1} at low Re (laminar). q_{c2} at high Re (turbulent). The maximum governs.

EQ 15 IEEE 738-2023	$q_r = 17.8 \times D \times \varepsilon \times [((T_c+273)/100)^4 - ((T_a+273)/100)^4]$	<i>Radiated heat loss [W/m]</i>
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Stefan-Boltzmann radiation from a cylinder. Constant $17.8 = \sigma \cdot \pi \cdot 10^8$ ($\sigma = 5.67 \times 10^{-8}$ W/m²·K⁴). Temperatures converted to Kelvin (+273). $\varepsilon \approx 0.5$ new, 0.8–0.9 weathered.

EQ 16 IEEE 738-2023	$q_s = \alpha_s \times Q_{se} \times \sin(\theta_{sun}) \times D$	<i>Solar heat gain [W/m]</i>
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Projected area per meter = D (diameter). $\sin(\theta_{sun}) = 1.0$ worst case (perpendicular sun). Egypt: $Q_{se} \approx 1000$ W/m².

EQ 17 IEEE 738-2023	$I_{max} = \sqrt{[(q_c + q_r - q_s) / R(T_{limit})]}$	<i>Maximum ampacity [A]</i>
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Heat balance rearranged to solve directly for current at $T_{limit} = 80$ °C (EEHC). No iteration required. Finding T from I requires Bisection on EQ 11 (R , q_c , q_r all depend on T simultaneously)

4. Equations Summary

EQ	Name	Standard	Formula	Purpose
EQ 1	Wind Pressure	IEC 60826	$q_0 = \frac{1}{2} \rho V^2$	Speed → pressure
EQ 2	Air Density	ICAO Std Atm	$\rho = \rho_0 \times (1 - 2.26e-5 \cdot H)^{5.26}$	Altitude correction
EQ 3	Span Factor G_L	IEC 60826 Annex B	$G_L = f(L)$	Reduce load: long spans
EQ 4	Wind Load /m	IEC 60826	$w_{wind} = q_0 \cdot C_{xc} \cdot G_c \cdot G_L \cdot d \cdot \sin^2 \Omega$	Corrected wind force
EQ 5	Resultant Load	IEC 60826	$w_r = \sqrt{(w_c^2 + w_w^2)}$	Vector combination
EQ 6	Parabolic Sag	Cable Theory	$sag = wL^2 / (8T)$	Mid-span sag
EQ 7	Swing Angle	IEC 60826	$\theta = \arctan(w_w / w_c)$	Deflection angle
EQ 8	Sag Components	IEC 60826	$v = sag \cdot \cos \theta$, $h = sag \cdot \sin \theta$	Clearance checks
EQ 9	State-Change	EN 50341	$T_2^3 - K \cdot T_2^2 - M = 0$	New tension (Newton-Raphson)
EQ 10	Safety Factor	EEHC	$T_z / RTS \leq 0.95$	Check vs. RTS
EQ 11	Heat Balance	IEEE 738	$q_c + q_r = q_s + I^2 R$	Master thermal eq.
EQ 12	Resistance $R(T)$	IEEE 738	$R_{20} [1 + \alpha_R (T - 20)]$	Temperature dependence
EQ 13	Air Properties	IEEE 738 Tables	ρ_f, μ_f, K_f at T_f	Film temperature
EQ 14	Convection q_c	IEEE 738	$MAX(q_{cn}, q_{c1}, q_{c2})$	Cooling by airflow
EQ 15	Radiation q_r	IEEE 738	$17.8 \cdot D \cdot \epsilon \cdot [...^4 - ...^4]$	Infrared emission
EQ 16	Solar gain q_s	IEEE 738	$\alpha_s \cdot Q_{se} \cdot \sin \theta \cdot D$	Sunlight heating
EQ 17	Ampacity	IEEE 738	$\sqrt{[(q_c + q_r - q_s) / R(T_{lim})]}$	Max current

5. Outputs

5.1 Mechanical Results

Output	Symbol	Unit	Description
Corrected wind load	w_{IEC}	kgf/m	Full IEC 60826 wind force per meter (EQ 4)
New tension	T_2	kgf	Conductor tension at analysis condition (EQ 9)
Safety factor	T_2/RTS	%	Must be $\leq 95\%$ per EEHC— red warning if exceeded
Total sag	sag	m	Maximum sag at mid-span (EQ 6)
Vertical sag	v_{sag}	m	Downward component for ground-clearance check
Horizontal swing	h_{swing}	m	Lateral component for structure-clearance check
Swing angle	θ	°	Conductor deflection from vertical (EQ 7)

5.2 Thermal Results

Output	Symbol	Unit	Description
Conductor temperature	T_c	°C	Steady-state temperature at operating current (EQ 11)
Temperature margin	$90 - T_c$	°C	Positive = spare capacity available
Ampacity at 90 °C	I_{max_90}	A	Maximum continuous current per EEHC (EQ 17)
Ampacity at 80 °C	I_{max_80}	A	Reference ampacity matching EEHC conductor tables
Joule heating	I^2R	W/m	Resistive loss at operating current
Convective loss	q_c	W/m	Air cooling — natural + forced (EQ 14)
Radiated loss	q_r	W/m	Infrared emission (EQ 15)
Solar gain	q_s	W/m	Sunlight absorbed (EQ 16)
Heat balance error	Δ	W/m	Residual — should be < 0.01 W/m if converged

5.3 Plot Tabs

Tab	Content
Mechanical — Sag & Swing	Vector diagram from support to mid-span. Dashed lines show vertical sag and horizontal swing. Title turns red if SF > 95%.
Thermal — IEEE 738	T _c vs. I curve 0 A to 120% I _{max} . Red dashed = 90 °C EEHC limit. Cyan dot = operating point. I _{max} annotated.
Wind Load Breakdown	Wind load vs. speed 0.5–50 m/s. Grey dashed = simple ($q_0 \times d$). Orange solid = full IEC 60826 corrected. Inset bar chart shows four IEC factors.

6. AL3 Conductor Database (EEHC Specification)

E = 6118 kgf/mm² (7-wire) · 5812 kgf/mm² (19-wire) · 5608 kgf/mm² (61-wire). $\alpha = 2.3 \times 10^{-5} / ^\circ\text{C}$ for all. CCC at 80 °C / 35 °C ambient / 0.6 m/s wind.

Conductor	d (mm)	w (kg/m)	A (mm ²)	RTS (kgf)	DC_R Ω/km	CCC (A)	Wires	E (kgf/mm ²)
16-AL3 (16 mm ²)	5.10	0.0434	15.9	478	2.0701	105	7	6118
24-AL3 (25 mm ²)	6.30	0.0662	24.2	729	1.3566	135	7	6118
34-AL3 (35 mm ²)	7.50	0.0938	34.0	1034	0.9572	170	7	6118
49-AL3 (50 mm ²) 7w	9.00	0.1351	49.5	1489	0.6647	210	7	6118
48-AL3 (50 mm ²) 19w	9.00	0.1327	48.3	1454	0.6841	210	19	5812
66-AL3 (70 mm ²)	10.50	0.1807	65.8	1979	0.5026	255	19	5812
S3-AL3 (95 mm ²)	12.50	0.2560	93.3	2805	0.3546	320	19	5812
117-AL3 (120 mm ²)	14.00	0.3212	117.0	3519	0.2827	385	19	5812
147-AL3 (150 mm ²)	15.80	0.4053	147.1	4426	0.2256	425	37	5812
182-AL3 (185 mm ²)	17.50	0.5003	181.6	5464	0.1827	490	37	5812
243-AL3 (240 mm ²)	20.30	0.6703	242.5	7296	0.1373	585	61	5608
299-AL3 (300 mm ²)	22.50	0.8275	299.4	9008	0.1112	670	61	5608
400-AL3 (400 mm ²)	26.00	1.1059	400.1	12037	0.0832	810	61	5608

7. EEHC Limits & Standards References

7.1 EEHC Environmental Design Limits

Parameter	Value	Applied in
Mean daily ambient temperature	35 °C	EDS reference temperature — base state for EQ 9
Maximum ambient temperature	50 °C	Extreme thermal scenario
Minimum ambient temperature	−10 °C	Max Swing load case (EQ 9, t_2)
Maximum conductor temperature	80 °C	Ampacity limit — T_{limit} in EQ 17
Maximum site altitude	1 000 m	ICAO density correction limit (EQ 2)
Maximum tension limit	95% RTS	Safety factor check (EQ 10) warning if exceeded

7.2 Standards References

Standard	Scope and Application
IEC 60826:2017	Design Criteria of Overhead Transmission Lines — wind loads, span reduction, combined load analysis. → EQ 1, 3, 4, 5, 7, 8.
IEEE 738-2023	Standard for Calculating the Current-Temperature Relationship of Bare Overhead Conductors — heat balance, air properties, convection, radiation, and ampacity. → EQ 11–17.
EN 50341	Overhead Electrical Lines Exceeding 1 kV AC — mechanical and structural design; basis for state-change cubic. → EQ 9.
ICAO Std Atmosphere	International Standard Atmosphere (ISO 2533) tropospheric air density vs. altitude model. → EQ 2.
EEHC Specification	Egyptian Electricity Holding Company — Technical Specification for AAAC/AL3 conductors: conductor database, environmental design limits. → EQ 10, all EEHC limits.